Dukaan

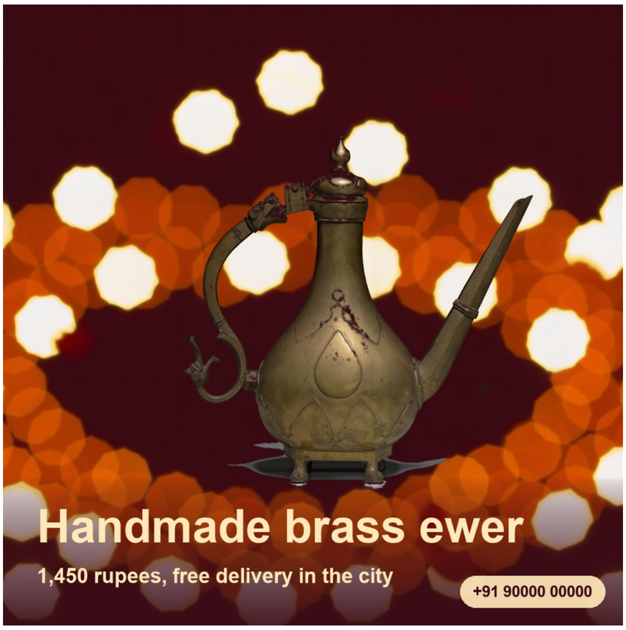
One product photo into three still-image advertising creatives, using AMD Radeon.

Track 1, Development of Multimodal Content Creation Tools himanshu748 | Radeon PRO gfx1100, 48 GB | ROCm 7.2.4 | LTX-2.3 22B

The problem

A shopkeeper can photograph stock quickly, but adapting one photo into legible feed, story, and banner layouts still takes design work. Generative tools can also change the product being advertised, so their output must be reviewed against the item the seller will actually ship.

What Dukaan does



One photo becomes a square for the feed, a 9:16 story, and a wide banner for a shop header. Candidate frames, a GIF preview, and optional audio can be retained for inspection and relabelling; they are not claimed as finished moving-media deliverables. The price and phone number are composited with Pillow after inference, never drawn by the model.

The GPU backend is invoked once per pack, not once per format. Three formats select different moments from the same candidate sequence. The optional refine setting performs an additional denoise pass inside that invocation.

One submitted photo across four fixed style presets:

The finding: the instance cannot run its own template

LTX-2.3 checkpoint	43 GB
Gemma 3 12B text encoder	23 GB
Total to load	**66 GB**
Container cap	**55 GB**

ComfyUI holds every model a graph touches for the life of the process. Loading both trips the cap and **the platform restarts the container mid-prompt**: JupyterLab returns with zero kernels, the port stops answering, and it presents as a network fault.

The fix: two processes that never overlap

phase	peak container RAM	wall
encode, text encoder only	35.4 GB	33 s
sample, checkpoint only	49.9 GB in the committed benchmark	about 55 s
stock template, one process	*trips 55 GB, restarts*	*n/a*

Peak becomes `max(43, 23)`, not `43 + 23`.

Plus: bf16 conditioning (it is read back while the checkpoint is resident), VRAM eviction before the VAE decode (the decode died with 2.13 GB free out of 47.98), and `torch.inference_mode()` around the phase (the LTX VAE updates the sampler's output in place).

Measured

setting	output	GPU	peak RAM
768x768, 49 frames	768x768	70.7 s	49.9 GB
768x768, refined	**1536x1536**	176.1 s	49.9 GB
3 products, one batch	768x768	**134.3 s**	49.9 GB

The committed runs held under the 55 GB cap. `dukaan bench` reruns the matrix on configured hardware; these historical measurements are not a performance guarantee.

Transport was a bottleneck too: 49 frames as 49 base64 requests took 3 m 34 s and reset the tunnel twice. One tar brought it to 2 m 19 s.

Batching is the lever that removes work rather than trading it

Loading the checkpoint costs 17 s and the text encoder 9 s, whatever you then generate. A shop packing fifty items one at a time pays that fifty times.

3 products, one batch	**134.3 s**
3 products, separately	212.1 s estimated baseline
per-product, across the batch	36.3 s, then 28.0 s, then 26.6 s

The 13.8 s load is paid once, and per-product time fell as the GPU warmed in the submitted batch. The 212.1-second comparison is calculated as three times the recorded 70.7-second single-product row; it is not three separately timed runs.

Three CPU decisions that keep GPU credits for the GPU

The cutout fits the background, it does not sample it. A quadratic surface per channel on the border ring, refit once with the worst residuals dropped. A plane left an elliptical pool of backdrop exactly where the product sits.

Stills are picked for spread, then sharpness. Contiguous windows, highest Laplacian variance in each. Frame 0 is never eligible: it is the plate the model was handed.

Reshaping mirrors, it does not crop or stretch. Cover-cropping a square to 9:16 cuts the product; clamping the edge row drew the bokeh into horizontal streaks.

37 tests, no GPU required. Without an instance configured, a deterministic CPU mock exercises command flow, file writing, layout, and selection. It does not validate LTX-2.3 quality, Radeon compatibility, product identity, or the historical benchmark. Generated creatives require review before publication. Apache-2.0. Demo photographs are CC0.